The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Abstract and storyline

Abstract

Temporal-interference (TI) stimulation applies two high-frequency currents whose superposition contains no field component at their low difference frequency. Population models generate that component through effective quadratic mixing and amplify it by recurrent resonance. For a smooth weak-input response with a nonzero second-order term, cellular nonlinearities contribute to a beat proportional to the product of the carrier amplitudes. A Jansen–Rit firing-rate sigmoid realizes this mixing through its curvature. An exact quadratic integrate-and-fire mean field provides a derived realization: its static rate curve emerges when membrane relaxation is fast relative to the input, while synapses remain dynamic. Outside that limit the mixing coefficient is frequency dependent. Fast cellular access and slow circuit resonance can therefore coexist, with membrane attenuation entering once, through access. Near a Hopf bifurcation, the network amplifies the generated beat as $1/\gamma$ within linear response, where γ is the modal damping. Coupling sweeps separate this gain from leading-order beat generation. Beat-locked rate modulation grows faster than the mean-rate change in the mean field and in a spiking network of 8000 neurons per population. In this weak-input regime, the carrier-product law makes an envelope-only dose map conditional on the transduction mechanism. TI and direct low-frequency stimulation share the amplifier when effective input coupling, readout, and operating state are matched, so resonance alone gives TI no efficiency advantage. The analysis concerns weak modulation of ongoing activity; biological access and the absolute response at human field strengths remain unresolved.

The abstract is the TN abstract of v0.14.6. K authorized the weak-field effective-response framing on 3 October 2026; the JNE build keeps a structured abstract with the same claims. Section and figure order remain unchanged from v0.14.4. The placement decisions on the last page remain open; this revision does not cut or reorder content.

Problem and theory

First argument block: Sections 1–2

A two-tone field carries no power at its beat frequency. A smooth weak-input response with a nonzero second-order term generates that frequency by mixing. Cellular contributions can be combined in an effective response, with QIF providing a derived realization. Access controls the input reaching the mixer; recurrent dynamics amplify its output. The static QIF curve is the adiabatic limit of its dynamic population response. Small amplitude alone does not remove memory or make its coefficient universal.

Place in v0.14.6	Scientific content to retain	Contribution to the argument	Examples / links
§1 Introduction	The two-tone field (Eq. 1) has lines only at the applied tones; a linear system cannot create the beat. Cellular mixing exists (MRG and Hodgkin–Huxley axon models; Luff's subthreshold mixer at 5 kHz, TTX-sensitive). Primate timing changes establish modulation; synaptic blockade and network simulations support circuit involvement. Retain weak modulation of ongoing activity as the scope, with direct stimulation thresholds and block outside it. Cellular contributions enter an effective quadratic response; access and amplification remain separate, without identifying an anatomical access route.	States the problem and the distinction the paper turns on: cellular mixing versus the network response it recruits. Announces the ladder (JR column, QIF mean field, spiking network, laminar column) once. The transfer curve must be defined before its curvature can mix, so §2.1 follows.	Fig. 1 (receiver concept). Grossman 2017; Mirzakhalili 2020; Luff 2024 (mixer); Vieira 2024; Caldas-Martinez 2024; CFC and HAM companion papers.
§2.1 Population transfer and the slow-synapse limit	JR synaptic filter and sigmoid (Eqs. 2–3); NMM1 with the direct field term $\Lambda[\hat{E}]$ (Eq. 3). The MPR mean field (Eq. 4) and its static curve $F(Q)$ (Eq. 5), inflection at $Q = \Delta/\sqrt{3}$, nonlinear even at zero recurrence and unbounded. Retain its derivation from the single-cell square-root rate and threshold recruitment. The slow-synapse reduction (Eq. 6) and its two conditions: $\tau_{\mathrm{rel}} \ll \tau_s$ makes the transfer static for synaptic input; $\omega_c \tau_{\mathrm{rel}} \ll 1$ is the stronger condition for a carrier. Effective curvature collects static mixing; the coefficient's value and dynamics remain model-specific. Cellular and circuit resonance are distinct; the reduction removes membrane-dependent modes.	Defines what a transfer curve is and when it is static, so that curvature can be a mixer. Introduces the time-scale distinction that the Discussion and Conclusion reuse. The field must reach the curve, which §2.2 supplies.	Rosetta Stone (Ruffini 2025); Montbrió 2015; Clusella 2023; de Palma Aristides 2026 (WP0242, unpublished); Hutcheon and Yarom 2000.

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§2.2 From applied field to a squared-envelope beat	Quasistatic field; polarization $s(t)$ through parallel entry points with access filters (Eqs. 7–8), one compartment response counted once. Taylor expansion (Eqs. 9–10) with the truncation condition $\|s\| \ll 1/\rho \approx 1.8$ mV. The square of the envelope (Eq. 11): A^2 is sinusoidal, A is a rectified cosine. The AM surrogate (Eq. 12), where it is used and where it is not. The effective dynamic coefficient $\chi^{(2)}$ combines cellular contributions and reduces to $\Sigma''/2$ in the static case. Retain smoothness and nonzero second-order response, its possible frequency dependence, and the DC term. Static predictions: ε^2 scaling, Σ'' law, carrier independence at fixed polarization.	Derives the beat and its amplitude: the first result. Fixes the waveform discipline (two tones for every waveform-sensitive claim). The derivation assumes a static curve, so §2.3 asks what a dynamic population does.	Figs. 3 and 6 (two-tone runs); Fig. 6d (surrogate check). HAM paper (Volterra kernel); Galan and Gadea 2023.
§2.3 The exact mean field connects static and dynamic mixing	Single-carrier DC coefficient $C(Q, \omega)$ (Eq. 13), with limits $F''/4$ and $F'/(2\omega^2)$ (Eq. 14), sign change at ω_z above the inflection. Fixed-excursion coefficient $K(\omega)$ (Eq. 15): the $1/\omega^2$ is the membrane, not a detector filter. The PING v_E^2 source matches the propagated second-order beat response to 1.1%, 0.1%, 0.03% at 300, 1000, 2000 Hz at four stable points. This is not a full finite-amplitude sweep validation. Carrier and beat take different routes to the rate (Table 1).	Connects the static and dynamic QIF descriptions, with membrane access counted once. Retain the distinction between the DC calculation and the two-tone beat source. Whatever the detector, its envelope law has observable signatures, which §2.4 derives.	Fig. 7; Table 1. Code: `static_vs_dynamic_check.py`, `mpr_twotone_lines_check.py`.
§2.4 Transduction class and its signatures	Cycle average (Eq. 16) is even in A; nonzero curvature gives a leading square law for a smooth memoryless transfer. Specified ideal rectifiers follow A; a step detector follows neither (harmonic ratio 0.151). Harmonic ladder 1/5, 3/35, 1/21 and r_{det} (Eq. 17). The network suppresses the second harmonic ($\mathcal{Q} = 10$: 20% reads 0.7%), so the test runs off resonance at $\Delta f \simeq f_0/4$, retaining the 1.22–1.25 transfer correction. Excursion map (Eq. 18): product law, 2× versus 40× example; not an account of focality. Spatial response (Eq. 19) and signed vector coupling.	Converts the detector class into measurements (amplitude slope, harmonic ratio, dose map) and separates spatial from dynamical selectivity. Generation gives a weak drive; §2.5 explains what makes it large.	App. D (bound and step detector); Fig. 6b (slope 1.99); Table 3 (predictions).
§2.5 Amplification: synaptic low-pass and network resonance	Synaptic transfer H_s (16 Hz corner; 0.7 at 10 Hz, 2.5% at 100 Hz). One-mode reduction, $\mu_\pm = -\gamma \pm i\omega_0$, $\mathcal{Q} = \omega_0/2\gamma$, gain $\propto 1/\gamma$ (Eq. 20); entrainment above the Hopf. Receiver equation (Eq. 21) with D and G and their normalization for JR and QIF. The cubic normal form has no quadratic mixer.	Supplies the second factor of the receiver and the claim that a large susceptibility is not evidence of detection. The carrier must still reach the mixer, which §2.6 quantifies.	Fig. 5; Figs. 4 and 9; App. C.

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§2.6 Carrier access depends on the effective membrane response	H_m (Eq. 22); $\lvert H_m\rvert^2 \approx 10^{-4}$ at 1 kHz for 16 ms, 0.39 and 0.14 at 1 and 2 kHz for 0.2 ms, labeled as assumed. Table 2 (response stages). AC space constant (Eq. 23), one tenth at 1 kHz. Channel kinetics belong in the effective mixing response, counted once; no universal gate cutoff follows from QIF. Carrier sweeps constrain access plus mixing; beat sweeps probe the amplifier. The plotted simulations retain finite membrane times; fast QIF relaxation is an analytic limit.	Separates carrier independence at fixed polarization from the applied-field roll-off and makes fast access a labeled hypothesis. The detector coefficient also depends on state, which §2.7 takes up.	Figs. S5, S6, S9; Deitcher 2017; Eyal 2016; Discussion paragraph on Cohen, Luff, Semenov.
§2.7 Operating-point dependence	PEIX (Eq. 24) tracks Σ' and peaks where Σ'' vanishes; the complementarity belongs to the memoryless detector. The fast-carrier QIF DC coefficient follows F'; a beat also needs its finite-frequency susceptibility. State shifts scale, and in JR can flip, the response. Retain the slope-versus-curvature distinction.	Turns the detector coefficient into a brain-state prediction and marks which part is model-specific. The theory is complete; Methods specify the simulations that test it.	Fig. 2b; Fig. 6a; CFC paper (PEIX).

JR, Jansen–Rit; MRG, McIntyre–Richardson–Grill; QIF, quadratic integrate-and-fire; MPR, Montbrió–Pazó–Roxin; NMM1/NMM2, sigmoid and exact mean-field neural mass models; PING, pyramidal–interneuron network gamma; AM, amplitude modulation; PEIX, Population Excitation–Inhibition Index; CFC, cross-frequency coupling; HAM, hierarchical amplitude modulation.

Simulations

Second argument block: Sections 3–4

The Results climb the ladder. The JR column verifies the three predictions of §2.2; the exact mean field reproduces generation and amplification with a derived nonlinearity; the spiking network confirms the timing signature at the microscale. Methods stand apart because the lock-in read-out and the pinned-operating-point protocol recur across models.

Place in v0.14.6	Scientific content to retain	Contribution to the argument	Examples / links
§3.1 Jansen–Rit: integration, lock-in, controls	Field enters only the pyramidal sigmoid; the read-out $v = y_1 - y_2$ is not fed the field. RK4 steps with $< 1\%$ convergence; settling and windows. Hopf at $p \approx 315$, $f_0 \approx 11.1$ Hz; SNIC at $p \approx 114$. Lock-in (Eq. 25) with DC subtraction to 10^{-8}. Open-loop detector versus closed-loop column; square law, linearization, finite-difference checks.	Makes every JR number reproducible and defines the lock-in that every later section uses.	Fig. 2; Grimbert and Faugeras 2006; `run_all.py`.
§3.2–3.4 LaNMM, NMM2 PING, QIF network	LaNMM: field as a signed polarization, not a clipped rate (a clip is a half-wave rectifier); alpha-power read-out; signed maps. PING (Eq. 27): weights $(1, 1, 1, 2)$, $\bar{\eta}_H = 1.110$, $f_0 = 54$ Hz, field as current $I_E = PE$ so the membrane computes access with $\tau_m = \tau_E$; two criticality knobs. QIF: $N = 8000$ per population, Lorentzian quantiles, beat 55 Hz (forced) and 42 Hz (gated, 11 Hz detuning); modulation depth and pairwise phase consistency; the 45% threshold sensitivity of phase concentration.	Fixes the models whose results follow and states which comparisons are perturbative. Decision for K: §3.2 serves only the App. E results and could travel with them.	Eqs. 26–27; Fig. S2 (LaNMM setup); Rosetta Stone PING parameters; Vinck 2010.
§3.5–3.6 Coupling sweeps, tACS control, vocabulary	Pinning: solve $p(C)$ for fixed v^*, read γ, ω_0 from the Jacobian; tACS control at $\varepsilon = 0.1$ mV. Three terms: forced response, entrainment (dominance screen $\lvert f_{\text{out}} - \Delta f \rvert < 0.3$ Hz), gating.	The pinning protocol isolates amplification from detection; the vocabulary keeps "entrainment" from covering gating.	App. F (metrics); Pikovsky 2001; Canolty 2010; Tort 2010.
§4.1–4.2 The sigmoid demodulates; the beat drives a resonance	An alpha line from two tones with no alpha power in the input. Lorentzian response growing $0.23 \to 0.43 \to 0.70$ mV toward the Hopf ($1/\gamma$); above the Hopf, Arnold-tongue entrainment read by output frequency.	The existence result: the beat is synthesized and amplified.	Figs. 3, 4; Fig. S4.
§4.3–4.4 Carrier independence, kHz roll-off, curvature law	Flat response over 25–300 Hz at fixed polarization; $1/f_c^2$ roll-off for 16 ms, retained access for 0.2 ms; two tones at 2 kHz with 2.2×10^{-13} input leakage. Curvature law to 0.9%; slope 1.99; linearization collapses the response 634×; two-tone/AM ratio 1.0000 at 0.02 mV and 0.988 at 1 mV.	Verifies the static detector predictions. Retain the one-point open-loop scope of the surrogate check; its extension to recurrent sweeps remains a review item.	Figs. 5, 6; Figs. S6, S9; Esmaeilpour 2020.

Place in v0.14.6	Scientific content to retain	Contribution to the argument	Examples / links
§4.5 Exact mean-field confirmation	PING generates and amplifies the beat through v_E^2; diagonal ridge with operating point; the detector sits on the fast side of its crossover, with a source operating-point independent to about 1%.	The receiver is not an artifact of the sigmoid.	Figs. 7, 8; Fig. S8; Clusella 2023.
§4.6 Coupling sets the gain at fixed detection	Pinned J-curve: $\sim 30\times$ growth, $1/\gamma$ then saturation; the same gain–γ relation via p; stable-focus side only. NMM2 J-curve without pinning.	The mesoscale claim: amplification depends on distance to criticality, not on which parameter delivers it.	Fig. 9; Fig. S10; Clusella 2023.
§4.7–4.8 Timing over rate; QIF confirmation	Modulation grows 13.5× (26 to 346 mHz) against a mean shift of 5.8× (−1.3 to −7.7 mHz); ratio 19 far from the Hopf, 45–66 near it. QIF: bunching below the Hopf; gating with sidebands at 95 and 11 Hz above it; modulation depth 0.003 → 0.29/0.23 with rate +8–10%; weak per-neuron locking.	Larger modulation than mean-rate change at mesoscale and microscale, qualitatively consistent with Vieira; retain the nonzero mean shift and weak per-neuron locking.	Fig. 10; Figs. S11, S12; Vieira 2024.
§4.9 Falsifiable predictions	Four predictions with tests: access versus mixing by carrier sweep; state and carrier dependence; transduction class by amplitude slope and harmonic ratio off resonance; frequency selectivity. Table 3 against existing evidence; synaptic block tests network involvement without excluding a cellular detector.	Converts the mechanism into falsifiable statements and bridges to the Discussion.	Table 3; Vieira 2024; Caldas-Martinez 2024; Opancar 2025.

LaNMM, laminar neural mass model; tACS, transcranial alternating-current stimulation; SNIC, saddle-node on an invariant circle; RK4, fourth-order Runge–Kutta.

Interpretation and supporting material

Third block: Sections 5–6 and Appendices A–G

The Discussion changes level from model results to physiology and states each scope limit at the claim it limits. The appendices hold the notation, the model equations, the two bounds, the laminar extension, and the metrics.

Place in v0.14.6	Scientific content to retain	Contribution to the argument	Examples / links
§5 Discussion	Weak-field cellular contributions enter an effective quadratic response; QIF provides a realization, without requiring channel identification. Weak amplitude and fast relaxation justify distinct approximations. Access candidates include nodes and AIS; strong terminal polarization alone does not establish fast charging (RC independent of area). Cifra excludes fields below 0.1 kV/m from the review's scope, not by a universal threshold. Cohen's periaxonal space ($\sim$ 1 ms fitted decay), Luff's 1 ms RC, Semenov's 0.7 μs charging: a 10–100$\times$ shorter time is a hypothesis. Dose map: min versus product and the DC term. Intensity changes the mechanism. Shared amplifier with tACS; temporal, not spatial, criticality. Couplings 0.12–0.5 mV per V/m; $\eta_{\mathrm{det}} \simeq |\Sigma''|\varepsilon/4\Sigma'$; $\mathcal{Q} \sim 10$. Vieira's 0.34 (0.04–0.65) against 0.8% and why that is not a rejection. The 0.348 supremum and its scope. Limitations: Plovie 2025, noise, columns, cascaded synapses cannot solve access, mixed human results.	Reinterprets each result in physiological terms and places every caveat where it bites. Retain all paragraphs; each adds a level (mechanism, time scales, anatomy, dosimetry, intensity, efficiency, scale, data, bound, limits). Retain matched-peak versus equal-RMS distinctions, and matching of input coupling, readout, and state for the shared amplifier.	Fig. S3 (tACS); Cohen 2020; Luff 2024 (PWM-TI); Semenov 2025; Vieira 2024; App. D.
§6 Conclusion	Effective quadratic mixing generates, resonance amplifies, and a large response does not identify a microscopic mixer. Static and dynamic descriptions connected by time scales; each filter once. Product law, shared amplifier, open access pathway.	Restates the storyline in three paragraphs without new claims.	—
App. A–B Notation and units; model equations	Table 4 (symbols, with the reused-glyph head note); JR equations (Eq. 28) and Table 5.	Recovery aids; definitions remain local.	—
App. C–D Cubic normal form; efficiency bound	Stuart–Landau has no second-order source (Eq. 31); numerical control at the floor. $\eta_{\mathrm{det}} \leq \rho\varepsilon/4$ (Eq. 32); family supremum 0.3479 at $\delta/2\varepsilon = 0.523$ (Eq. 34), a step detector with harmonic ratio 0.151; the microvolt width is an idealization, and the maximum is numerical, not certified.	C supports §2.5 (susceptibility is not detection); D supports §2.4 and the Discussion's efficiency paragraphs.	`threshold_bound_check.py`.

Place in v0.14.6	Scientific content to retain	Contribution to the argument	Examples / links
App. E Laminar column at low carriers	Tongues at 25–80 Hz carriers up to 3 mV, outside the perturbative regime; 95%/97% of the P_1-driven grids below baseline; double resonance under superficial drive (7% off resonance) against broadband response under deep drive (25%); three LaNMM predictions, the carrier-matching one untested.	Shows a band-limited intermediary makes the carrier choice matter; does not carry the kHz argument.	Fig. S1, S2, S7; Gordon 2019.
App. F–G Vocabulary metrics; supplementary figures	Dominance screen versus capture test; folded-rate metric; Arnold-tongue protocol. Figures S2–S12 support §4.2–4.8 and the Discussion.	Metrics behind the three terms; the figure bank for the JNE supplement.	Figs. S2–S12.

Placement decisions for K. (1) Keep the seven Theory subsections in place, or move some detailed interpretation after Results. The review recommends moving §2.4's detailed transduction tests while retaining carrier access (§2.6) before its Results application. (2) Keep LaNMM Methods §3.2, or move it beside its results in App. E. (3) Promote S3 (tACS), S10 (NMM2 J-curve), S11 (modulation versus mean rate), or none; the review recommends S11 first and retaining all 22 figures. None of these placement changes has been executed. K authorized the weak-field framing and the corresponding abstract revision in v0.14.5. Remaining claim and reproduction issues are recorded in `docs/review_2026-10-03_storyline_r1_r13.md`; frequency-dominance maps do not establish phase capture, and the LaNMM lock-in percentages need their supporting calculation.